Spring Boot Error Notes

ClassCastException in @RestControllerAdvice | Easy Language Guide

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1. The Error — In Simple Words

> **Error in Plain English:** Java said: "You gave me a complex object (ApiResponse), but I was expecting raw bytes (byte[]). I cannot convert one into the other!"

Think of it like this: you ordered a pizza box, but someone put a full dining table inside it. The delivery guy cannot carry that — wrong container!

2. Key Terms Explained

Keyword	Simple Meaning	In Our Case
@RestControllerAdvice	A global interceptor — sits between your controller and the response sent to client	`GlobalResponseHandler class`
ResponseBodyAdvice	An interface that lets you modify the response body before it is sent	`Used to wrap every response in ApiResponse`
ApiResponse	Your custom wrapper object that holds status, message, and data	`com.ankit.module4.advice.ApiResponse`

byte[] / [B	Raw bytes — binary data like images, files, YAML. [B is Java JVM name for byte[]	`Swagger UI serves .yaml files as byte[]`
HttpMessageConverter	Spring's tool that converts Java objects into HTTP response format (JSON, bytes, etc.)	`ByteArrayHttpMessageConverter vs Jackson`
ByteArrayHttpMessageConverter	Handles byte[] responses — writes raw binary to client	`Was wrongly applied to ApiResponse`
MappingJackson2HttpMessageConverter	Converts Java objects to JSON — this is what ApiResponse needs	`Should have been used instead`
ClassCastException	Java error when you try to treat one type of object as a different incompatible type	`ApiResponse treated as byte[]`
basePackages	Limits which controllers the advice applies to	`Should be set to com.ankit.module4`
DevTools RestartClassLoader	Spring Boot's hot-reload classloader — loads classes separately, can cause	`Made the class mismatch worse`

	identity issues	
springdoc / Swagger	Library that generates API documentation UI — serves raw files internally	`Its endpoints were intercepted by mistake`
Actuator	Spring Boot's built-in health/metrics endpoints — also serves binary data at times	`Also affected by the unscoped advice`

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3. How the Error Happened — Step by Step

Follow this flow to understand exactly what went wrong:

Step 1	**Swagger UI is called** A request hits a Swagger/Actuator endpoint that is supposed to return a raw .yaml file or binary icon — its return type is byte[].
Step 2	**Spring picks a converter** Spring looks at the return type byte[] and assigns ByteArrayHttpMessageConverter to write the response. This is correct so far.
Step 3	**Your advice intercepts** Before the response is sent, Spring passes the byte[] through your GlobalResponseHandler. Because basePackages was not set, it intercepts ALL endpoints — including Swagger.
Step 4	**Wrapping happens** Your advice wraps the byte[] inside a new ApiResponse object. Now instead of byte[], Spring has an ApiResponse object to send.

Step 5	**Converter tries to write** Spring still has ByteArrayHttpMessageConverter assigned from Step 2. It tries to cast ApiResponse to byte[] to write it. CRASH — ClassCastException is thrown.

4. The Root Cause — One Sentence

Root Cause: Your @RestControllerAdvice had no basePackages filter, so it intercepted Swagger and Actuator endpoints that return byte[], and tried to wrap them in ApiResponse — but Spring had already assigned ByteArrayHttpMessageConverter for those responses.

5. The Fixes

Fix 1 — Scope your Advice (Recommended)

```
@RestControllerAdvice(basePackages = "com.ankit.module4")
```

✅ **Why it works:** This is the cleanest fix. It tells Spring: only intercept controllers inside MY package. Swagger and Actuator live in different packages, so they are automatically excluded.

Fix 2 — Guard inside beforeBodyWrite()

```
if (body instanceof byte[]) return body;  // skip raw bytes if (body instanceof ApiResponse) return body;  // skip already-wrapped
```

✅ **Why it works:** Even if your advice intercepts byte[] responses, it checks the type first and returns it unchanged. A safety net.

Fix 3 — Filter in supports() method

```
return !className.startsWith("org.springdoc") &&
!className.startsWith("org.springframework.boot.actuate");
```

✅ **Why it works:** The supports() method is called before beforeBodyWrite(). If it returns false, the advice is skipped entirely for that request.

Fix Comparison

Fix	What it does	When to use
basePackages	Scopes the advice to your package only — proactive prevention	Best default choice
byte[] check	Defensive guard — skips wrapping for raw binary responses	Good safety net to add
supports() guard	Explicitly excludes third-party libraries from interception	Use if others are affected too

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6. Quick Summary Cheat Sheet

What happened?	Swagger endpoint returned byte[], your advice wrapped it in ApiResponse, Spring's ByteArrayHttpMessageConverter crashed trying to cast it
Why did it intercept Swagger?	No basePackages defined — advice applied to every controller in the app

What is [B in Java?	JVM notation for byte[] (raw byte array)
What is ClassCastException?	Java error when you try to use an object as a wrong type
Best fix?	Add basePackages = "com.ankit.module4" to @RestControllerAdvice
Extra safety?	Add instanceof byte[] check inside beforeBodyWrite()

Remember: always scope your @RestControllerAdvice to your own package!